

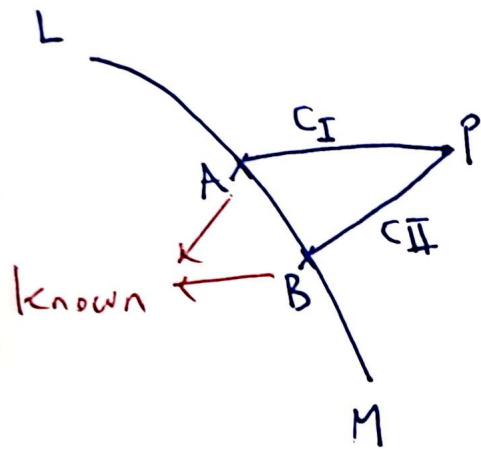
Method of characteristics

What we learned so far? (Good for Q1-HW#10)

If we have a hyperbolic Eqⁿ with n -unknowns, after we add $(n-1)$ helping Eqⁿ, and form $n \times n$ matrix of coefficients, then we use Kramer's rule & solve $n \times n$ system, put $|A|$ (determinant of denominator) $= 0$, find charac. lines, which are $(n-1)$ lines, then put numerator $= 0$.
→ find compatibility Eqⁿ

We did so & find for compressible potential flow Eqⁿ

$$\begin{cases} \frac{dy}{dx} = \tan(\alpha \mp \mu) \\ v \pm \alpha = \text{cte} \end{cases}$$

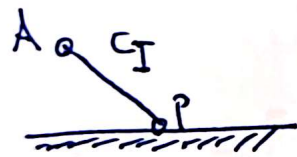


interior of the domain

$$C_I = \overbrace{v_A + \alpha_A}^{\text{known}} = \underbrace{v_P + \alpha_P}_{\text{unknown}}$$

$$C_{II} = \underbrace{v_B - \alpha_B}_{\text{known}} = \underbrace{v_P - \alpha_P}_{\text{unknown}}$$

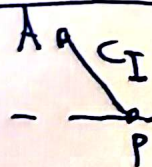
$$\text{Solve: } \begin{cases} v_P = \frac{(v_A + \alpha_A) + (v_B - \alpha_B)}{2} \\ \alpha_P = \frac{(v_A + \alpha_A) - (v_B - \alpha_B)}{2} \end{cases}$$



wall

We know $\alpha_P = \alpha_{\text{wall}} = \text{known}$

$$C_I = \underbrace{v_A + \alpha_A}_{\text{known}} = \underbrace{v_P + \alpha_P}_{\text{known}} \rightarrow \text{we found } v_P \xrightarrow{\text{P.M.}} M_P \text{ of } \mu$$



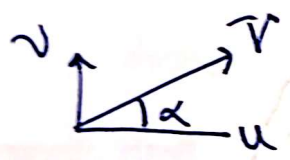
free-stream

We know M_P (or v_P)

$$C_I = \underbrace{v_A + \alpha_A}_{\text{known}} = \underbrace{v_P}_{\text{known}} + \underbrace{\alpha_P}_{\text{is found!}}$$

So, instead of solving complex PDE of potential, we solve
 charac. Eqⁿ & compatibility Eqⁿ.

for supersonic ~~com~~ potential flow, we have:

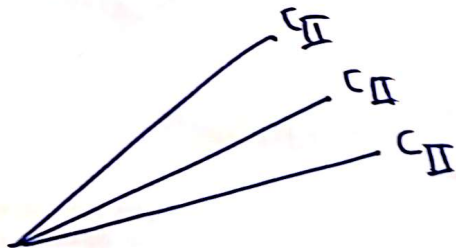
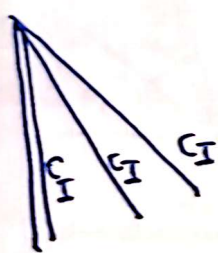


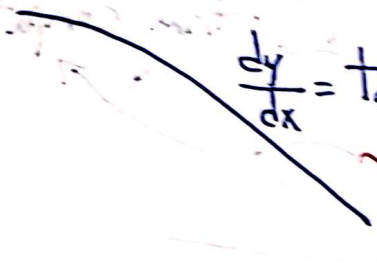
α : flow angle
 μ : Mach wave angle

$$\frac{dy}{dx} = \tan(\alpha + \mu)$$

$v - \alpha = cte = c_{II}$

Why c_{II} goes upward $\rightarrow \frac{dy}{dx} = \tan(\alpha + \mu) > 0$





$$\frac{dy}{dx} = \tan(\alpha - \mu) \text{ (usually } < 0 \text{)}$$

$v + \alpha = cte = c_I$

the aim is to know nozzle profile such that no-wave is reflected from the walls.

❖ By working outward from the initial value curve LM, it is then possible to build a net or mesh, with points connected by the characteristics.

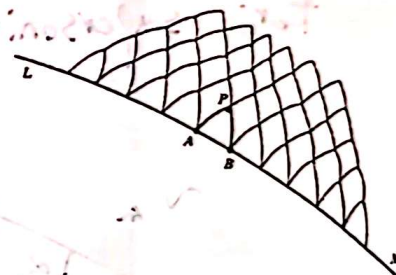
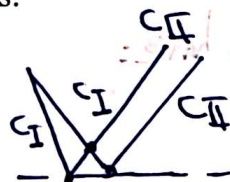


Figure 14.7 Characteristic Mesh

❖ Direction of characteristics:

$$\left(\frac{dy}{dx}\right)_I = \tan(\alpha - \mu)$$

Finite difference version

$$\left(\frac{dy}{dx}\right)_{II} = \tan(\alpha + \mu)$$

$$\left(\frac{dy}{dx}\right)_I = \frac{y_P - y_A}{x_P - x_A} = \tan(\alpha - \mu)_A = m_I$$

$$\left(\frac{dy}{dx}\right)_{II} = \frac{y_P - y_B}{x_P - x_B} = \tan(\alpha + \mu)_B = m_{II}$$

two Eq^s for two unknowns x_P & y_P

✓ improved calculation:

$$\left(\frac{dy}{dx}\right)_I = \frac{y_P - y_A}{x_P - x_A} = \tan\left[\frac{(\alpha - \mu)_A + (\alpha - \mu)_P}{2}\right] = m_I$$

$$\left(\frac{dy}{dx}\right)_{II} = \frac{y_P - y_B}{x_P - x_B} = \tan\left[\frac{(\alpha - \mu)_B + (\alpha - \mu)_P}{2}\right] = m_{II}$$

from compatibility Eq^s

$$y_P = y_A + m_I(x_P - x_A)$$

$$y_P = y_B + m_{II}(x_P - x_B)$$

$$x_P = \frac{y_A - y_B + m_{II}x_B - m_Ix_A}{m_{II} - m_I}$$

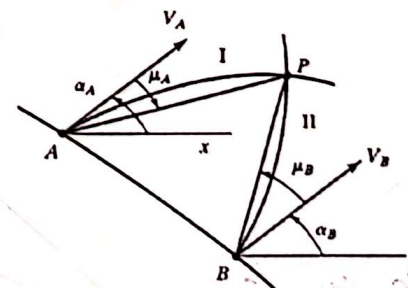
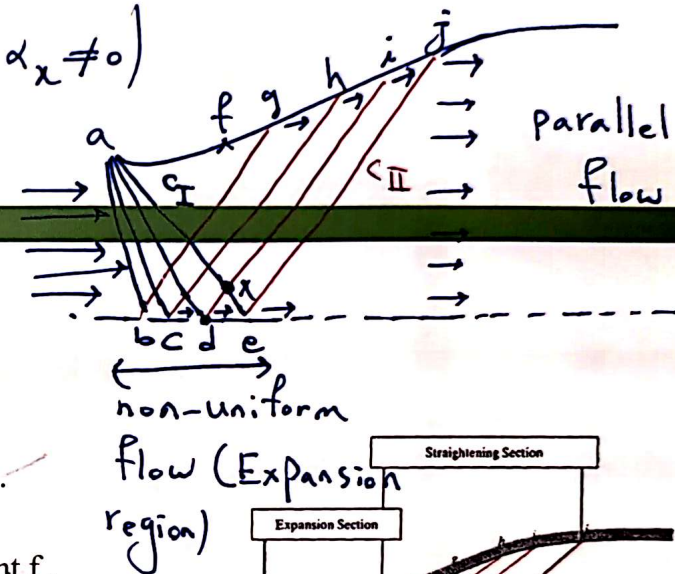


Figure 14.8 Straight-Line Approximations of the Characteristics

$$(d-i) C_{II}: \cancel{v_d - \alpha_d} = v_x - \alpha_x = v_i - \alpha_i \quad (\alpha_i \neq \alpha_x \neq 0)$$

$$(e-j) C_{II}: \cancel{v_e - \alpha_e} = v_j - \alpha_j \Rightarrow \alpha_j = 0$$



❖ A converging-diverging supersonic nozzle:

- Sonic flow at the nozzle throat.
- Consider only one half of the flow (the centerline is a streamline).
- The wall angle increases from point a to a maximum value at point f.
- Four expansion waves (Type I characteristics) in the expansion section of the nozzle intersect the centerline at points b, c, d, and e and are reflected, as characteristics of Type II, impinging on the nozzle wall at points g, h, i, and j, respectively.
- The waves reflected from the centerline turn the flow back to the horizontal between points e and j.
- The wall contours at points g, h, i, and j must turn the flow through the same angle as the incident waves so as to cancel those waves.

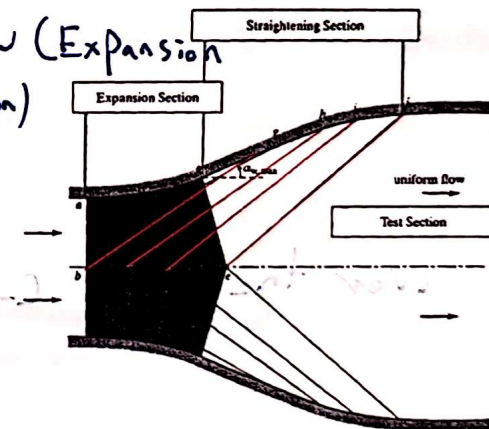
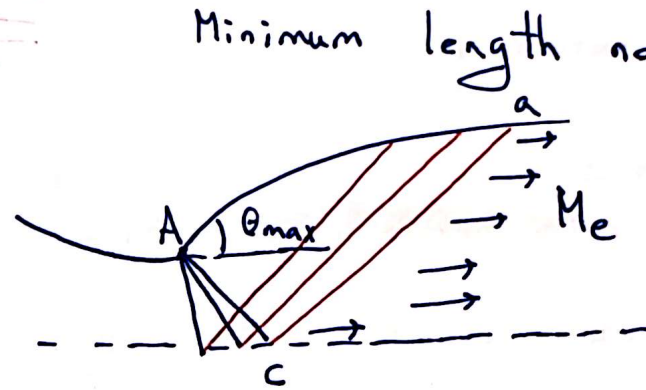
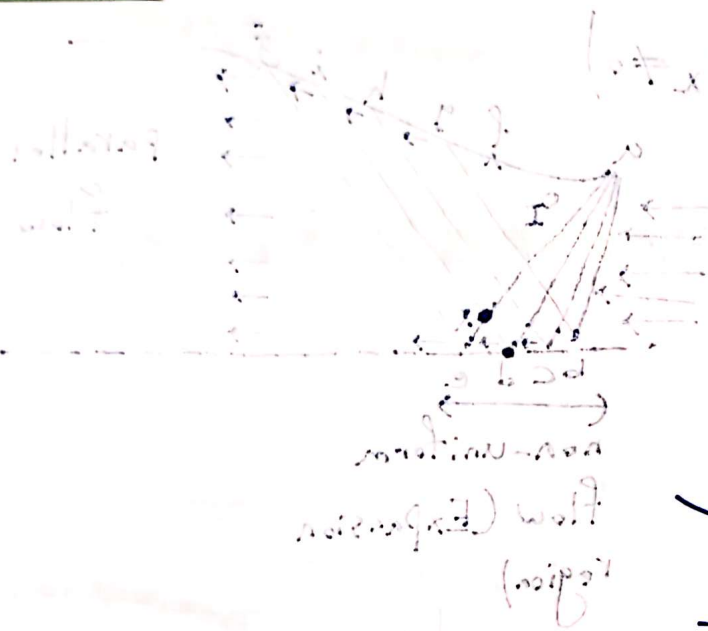


Figure 14.25 Wave Pattern in a Supersonic Nozzle Containing Both Expansion and Straightening Sections

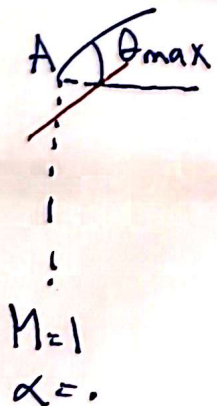


over the line C-a

$$C_{II} \rightarrow v_c - \alpha_c = v_a - \alpha_a$$

$$\alpha_c = 0 \implies \alpha_a = 0 \implies v_c = v_a = v_{Me}$$

" " " A-C $C_I \rightarrow v_A + \alpha_A = v_c + \alpha_c = v_{Me}$ (1)



P-M function between $M=1$ & M after point A

$$\left\{ \begin{array}{l} \theta_{max} - \theta_{@M=1} = v_A - v(M=1) \\ \rightarrow \theta_{max} = \alpha_A = v_A \end{array} \right. \quad (2)$$

$$\Rightarrow \theta_{max} = v_{Me/2}$$